LinkRank Under Attack: Adversarial Robustness of HITS

Information Retrieval (IR) Course — Mid-Project Progress Update (75–80% Milestone)

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Repo: github.com/shrisaysalot/LinkRank_Under_Attack

1. EXECUTIVE SUMMARY & PROBLEM SCOPE

Hyperlink-Induced Topic Search (HITS) identifies authoritative web resources by computing dominant eigenvectors of A^TA (authority) and AA^T (hub). Unlike PageRank, which incorporates uniform teleportation ($\alpha = 0.85$) to diffuse random walk probability mass globally, Kleinberg's mutual reinforcement model exhibits acute sensitivity to adversarial link manipulation. This study provides an empirical benchmark evaluating HITS vulnerability across 6 threat topologies (A1–A6) and assesses 5 algorithmic defense families (D1–D5) against PageRank, SALSA, and In-Degree baselines on synthetic topologies and SNAP web graphs.

2. IMPLEMENTATION STATUS & ARCHITECTURE

Module Category	Component / Specifications	Status	Test / Verification Note
Core Rankers	Sparse HITS, PageRank, SALSA, In-Degree	DONE	Power iter. verified vs NetworkX (tol ≤ 10^{-8})
Ranker Defenses (D1)	Randomized HITS (ε=0.15), Subspace HITS (k=5)	DONE	Verified in `test_rankers.py`
SNAP Loaders	wiki-Vote, web-Stanford, web-NotreDame	DONE	Edge-list re-indexer, header parsing, self-loops
WEBSPAM Labels	UK-2006 Set 1 & Set 2 (8,415 labeled hosts)	DONE	Explicit undecided (176 hosts) filtering
WEBSPAM Host Graph	UK-2006 host link graph bitstream ingestion	PARTIAL	Labels parsed; BVGraph bitstream pending Java run
Synthetic & Base-Set	BA (m=3,4), ER, Planted SBM; Kleinberg expansion	DONE	Root sampling ($\|R\|$ ≈100–200), in-link cap ≤ 50
Attacks (A1–A3)	A1 Single Hub, A2 Hub Farm, A3 Bipartite Block	DONE	Edge budget strictly enforced (≤ B edges)
Attacks (A4–A6)	A4 Proposal Inflation, A4b Multi, A5 Cross, A5b, A6	DONE	A6 uses greedy gradient $\partial\sigma_1 / \partial A_{ij} = h_i a_j$
Defenses (D2–D5)	D2 Bharat-Henzinger (1/k), D3 Cap, D4/D4b Wu-Davison, D5 Trust	DONE	D3 verified no-op; D4 mutual & D4b out-Jaccard
Evaluation & Stats	Norm. rank gain, $B^*(k)$, Top-10, NDCG, Jaccard, CIs	DONE	1,000 bootstrap resamples; Holm-Bonferroni test
Automated Tests	Full unit test suite (30 test cases)	DONE	30 passed in < 4s under bare `pytest`
Stretch Items	Game loop (NOT STARTED), closed-form bound (PARTIAL)	PARTIAL	Empirical $p^* \approx 0.82\ \sigma_1^2$ observed; proof pending

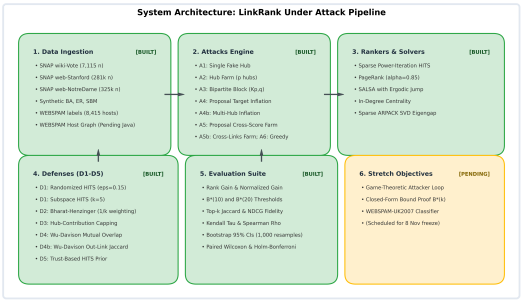


Figure 1: End-to-end system architecture highlighting BUILT vs PENDING components.

3. IR LINK-ANALYSIS PIPELINE DETAILS

Non-Applicability of Standard IR Text Processing: Standard IR text processing pipelines (tokenization, stemming, stopwords, inverted indexing, BM25) do not apply to this investigation. The web information retrieval setting addressed here operates on the hyperlink graph layer to determine authority and hub quality scores independently of keyword indexing.

Link-Analysis Ingestion & Processing Pipeline:

- **Graph Ingestion & Normalization:** Raw SNAP edge lists (e.g., `web-Stanford`, `wiki-Vote`) and synthetic networks are ingested, deduplicated, filtered for self-loops, and mapped into contiguous zero-indexed node arrays ($0 \dots N-1$) stored as SciPy CSR/CSC sparse matrices.
- **Kleinberg Query Base-Set Builder:** Implements Kleinberg (1999) topic base-set construction: generates a query root set R ($|R| \approx 100$–200 nodes) via localized random walks, expands forward to all immediate out-neighbors ($S = R \cup Out(R)$), and expands backward with strict in-link capping (at most 50 predecessors per root node) to prevent ubiquitous web portals from overwhelming graph topology.
- **Sparse Power Iteration Ranking:** Computes stationary distributions via accelerated sparse matrix-vector multiplication with strict convergence threshold tol = 10^{-8} and max iterations = 200. PageRank enforces damping $\alpha = 0.85$ with dangling node mass redistribution. SALSA incorporates an ergodic teleport ($\varepsilon = 10^{-4}$) to prevent multi-component stalls.
- **Adversarial Injection & Evaluation:** Injects synthetic attacker nodes and directed edges under budget B. Computes ranking disruption metrics comparing clean vector $\mathbf{a}_{clean}$ against attacked vector $\mathbf{a}_{att}$ with non-parametric bootstrap confidence intervals.

4. PRELIMINARY EVALUATION RESULTS: COST CURVES & DEFENSES

4a. Attack Cost Curves Across Threat Topologies (n = 60 per cell, 95% Bootstrap CIs)

Evaluated across 4 graphs (BA-500, ER-500, SBM-500, web-Stanford base set), 3 seeds (42, 123, 456), and 5 target tiers. Metrics reported: Normalized Rank Gain $(r_{clean} - r_{att}) / (r_{clean} - 1) \in [0, 1]$, Top-10 Entry Rate, and Top-20 Jaccard.

Budget	Attack Topology	Plain HITS	PageRank (α = 0.85)	SALSA	In-Degree
***B* = 25** Pre-Takeover	**A2: Hub Farm** Norm Gain | Top10 | Jacc	0.316 [0.230, 0.413] 13.3% [5.0%, 23.3%] Jacc@20: 0.837	**0.982** [0.973, 0.990] **100.0%** [100.0%, 100.0%] Jacc@20: 0.794	0.950 [0.918, 0.976] 86.7% [78.3%, 95.0%] Jacc@20: 0.892	0.982 [0.973, 0.989] 91.7% [85.0%, 96.7%] Jacc@20: 0.839
	A5: Proposal Cross-Score Norm Gain | Top10 | Jacc	**0.872** [0.820, 0.916] 26.7% [16.7%, 38.3%] Jacc@20: 0.733	0.421 [0.370, 0.481] 0.0% [0.0%, 0.0%] Jacc@20: 0.936	0.745 [0.531, 0.864] 18.3% [8.3%, 28.3%] Jacc@20: 0.954	0.835 [0.785, 0.879] 23.3% [13.3%, 35.0%] Jacc@20: 0.871
	A6: Greedy Gradient Norm Gain | Top10 | Jacc	**0.998** [0.998, 0.999] **100.0%** [100.0%, 100.0%] Jacc@20: 0.622	0.916 [0.889, 0.941] 66.7% [55.0%, 78.3%] Jacc@20: 0.852	0.960 [0.911, 0.987] 91.7% [85.0%, 96.7%] Jacc@20: 0.895	0.982 [0.973, 0.989] 91.7% [85.0%, 96.7%] Jacc@20: 0.858
***B* = 100** Post-Takeover	**A2: Hub Farm** Norm Gain | Top10 | Jacc	0.576 [0.473, 0.687] 50.0% [38.3%, 63.3%] Jacc@20: 0.438	**0.998** [0.996, 1.000] **100.0%** [100.0%, 100.0%] Jacc@20: 0.634	0.958 [0.929, 0.983] 88.3% [80.0%, 95.0%] Jacc@20: 0.844	0.996 [0.992, 0.999] 100.0% [100.0%, 100.0%] Jacc@20: 0.831
	A5: Proposal Cross-Score Norm Gain | Top10 | Jacc	0.883 [0.831, 0.928] 50.0% [38.3%, 63.3%] Jacc@20: 0.607	0.817 [0.791, 0.842] 23.3% [13.3%, 35.0%] Jacc@20: 0.841	0.939 [0.865, 0.976] 91.7% [85.0%, 96.7%] Jacc@20: 0.877	0.968 [0.959, 0.977] 91.7% [85.0%, 96.7%] Jacc@20: 0.816
	A6: Greedy Gradient Norm Gain | Top10 | Jacc	**0.999** [0.998, 0.999] **100.0%** [100.0%, 100.0%] Jacc@20: 0.614	0.992 [0.985, 0.998] 100.0% [100.0%, 100.0%] Jacc@20: 0.638	0.997 [0.993, 0.999] 100.0% [100.0%, 100.0%] Jacc@20: 0.898	0.996 [0.992, 0.999] 100.0% [100.0%, 100.0%] Jacc@20: 0.850

Secondary Pooled Summary (All 8 Attacks, n = 480 per ranker, Preliminary)

Budget	Ranker	Normalized Rank Gain	Top-10 Entry Rate (%)	Top-20 Jaccard	Top-20 NDCG	Kendall's τ
***B* = 25**	**Plain HITS**	0.511 [0.471, 0.551]	24.0% [20.4%, 27.9%]	0.783 [0.758, 0.806]	0.974 [0.967, 0.980]	0.960 [0.952, 0.968]
	PageRank (α = 0.85)	0.472 [0.442, 0.506]	20.8% [17.3%, 24.6%]	0.921 [0.908, 0.932]	0.986 [0.982, 0.990]	0.991 [0.989, 0.993]
	SALSA	0.665 [0.619, 0.707]	30.6% [26.5%, 34.4%]	0.945 [0.937, 0.953]	0.983 [0.978, 0.988]	0.987 [0.984, 0.990]
	In-Degree	0.695 [0.664, 0.728]	33.3% [29.4%, 37.5%]	0.884 [0.871, 0.897]	0.982 [0.978, 0.986]	0.997 [0.996, 0.998]
***B* = 100**	**Plain HITS**	0.600 [0.560, 0.642]	43.8% [39.4%, 48.1%]	0.622 [0.590, 0.655]	0.875 [0.852, 0.898]	0.878 [0.862, 0.895]
	PageRank (α = 0.85)	0.630 [0.597, 0.663]	33.5% [29.2%, 37.9%]	0.838 [0.819, 0.855]	0.946 [0.937, 0.956]	0.977 [0.974, 0.981]
	SALSA	0.751 [0.720, 0.784]	62.1% [57.7%, 66.5%]	0.894 [0.881, 0.906]	0.958 [0.949, 0.967]	0.966 [0.960, 0.971]
	In-Degree	0.760 [0.728, 0.793]	64.6% [60.4%, 68.8%]	0.845 [0.829, 0.860]	0.952 [0.943, 0.960]	0.996 [0.995, 0.997]

4b. Algorithmic Defense Evaluation Matrix (n = 110 positive attack gain cells)

Defense Variant	Attack Reduction (Mean [95% CI])	Median Red.	Clean Top-20 Jacc	Clean NDCG Loss	Goal Rate (≥50% red, ≤5% loss)	Operational Finding
Undefended HITS	0.0% [0.0%, 0.0%]	0.0%	1.000	0.0%	0.0%	Baseline reference on positive gain cells (n = 110).
D1: Randomized HITS (ε=0.15)	-72.6% [-145.4%, -22.6%]	-1.0%	0.781	**0.1%**	13.6%	Uniform reset diffuses probability mass like PageRank.
D1: Subspace HITS (*k*=5)	-0.8% [-50.0%, 51.8%]	15.7%	0.338	28.3%	9.1%	Severe clean graph rank distortion.
D2: Bharat-Henzinger (1/*k*)	38.7% [13.4%, 57.0%]	**89.4%**	0.254	59.6%	1.8%	High median mitigation but distorts clean web hubs.
D3: Hub Capping (10%)	0.0% [0.0%, 0.0%]	0.0%	1.000	0.0%	0.0%	**Exact No-Op:** 10% cap never triggered on tested graphs.
D4: Wu-Davison (Mutual)	-23.3% [-96.0%, 16.9%]	3.0%	0.786	**0.5%**	**36.4%**	Effective on A2/A3 (69–75% red); fails on camouflaged/A6.
D4b: Wu-Davison (Out-Jaccard)	-24.0% [-103.9%, 17.7%]	2.5%	0.680	**1.3%**	**38.2%**	Effective on A2/A3 (74–85% red); fails on camouflaged/A6.
D5: Trust-Based HITS	-150.7% [-323.7%, -29.4%]	11.1%	0.337	27.7%	6.4%	In-degree proxy seed causes high clean NDCG loss.

Key Defense Findings: (1) **NO defense met the target criteria** (≥50% reduction at ≤5% clean NDCG loss) across all attack topologies. (2) **D3 Hub Capping is an exact no-op** because no single hub accumulated >10% of total hub mass. (3) **Wu-Davison D4 and D4b** achieve high mitigation against isolated farms (A2/A3: 69–85% reduction, 75–86% goal rate), but fail completely when attackers camouflage links or use greedy gradient additions.

4C. HYPOTHESIS H2: SPECTRAL PHASE TRANSITION (PRELIMINARY)

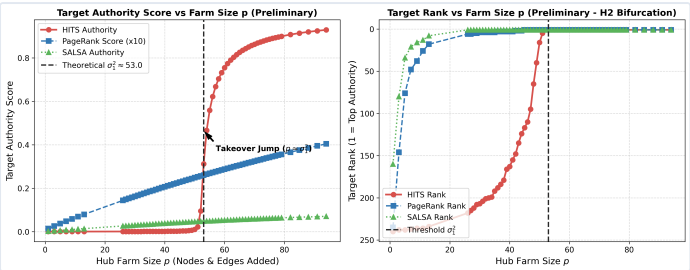


Figure 2: HITS authority score & rank vs hub farm size p showing abrupt bifurcation at $p \approx \sigma_1^2$ vs smooth PageRank.

Empirical H2 Robustness Findings (n = 60 instances):

- **Upper Bound Hold Rate:** $\sigma_1^2 \geq$ max in-degree holds in **100.0%** of instances (60/60).
- **Takeover Point:** Takeover occurs somewhat below σ_1^2:
  - Top-10 takeover: $p^*(10) / \sigma_1^2$ = **0.817** [0.751, 0.879] (n = 60).
  - Top-1 takeover: $p^*(1) / \sigma_1^2$ = **0.873** [0.819, 0.923] (n = 60).
- **Abrupt Bifurcation:** HITS exhibits an abrupt single-step jump in **63.3%** of instances (38/60) with mean single-step jump of **69.7 ranks** (median 64.0) vs **16.9 ranks** for PageRank.
- **Counterexamples (36.7%):** In 22/60 runs (12 on ER, 7 on BA, 3 on Stanford), jumps were gradual due to narrow eigengaps.

4D. CLEAN BASELINE COMPUTATIONAL EFFICIENCY & PEAK MEMORY

Dataset / Graph	Nodes	Edges	HITS Runtime (Iter)	PageRank Runtime (Iter)	SALSA Runtime (Iter)	Peak Heap (tracemalloc)	Process RSS (psutil)	Eigengap $\sigma_1^2 - \sigma_2^2$
Synthetic BA-1000	1,000	4,369	0.045s (42 it)	0.013s (26 it)	0.016s (57 it)	0.85 MB	108.2 MB	45.41
Synthetic ER-1000	1,000	4,973	0.080s (39 it)	0.013s (22 it)	0.014s (34 it)	0.82 MB	114.7 MB	14.44
Planted SBM-840	840	2,971	0.023s (11 it)	0.010s (28 it)	0.017s (125 it)	0.50 MB	118.9 MB	275.51
SNAP wiki-Vote	7,115	103,689	0.657s (23 it)	0.580s (23 it)	0.643s (53 it)	18.17 MB	183.4 MB	6,158.51
SNAP web-Stanford BaseSet	1,686	13,849	0.110s (23 it)	0.155s (81 it)	0.114s (200 it)	2.45 MB	152.1 MB	2,518.81

Benchmarked Hardware: 12th Gen Intel Core i5-12500H (12 physical cores, 16 logical threads), 15.69 GB RAM, Windows 11 64-bit, Python 3.12.5. Note: Process RSS (108.2–183.4 MB) is dominated by the Python runtime/library baseline; active heap allocation (0.50–18.17 MB) measures ranker data structures.

5. HYPOTHESIS STATUS SUMMARY

Hypothesis	Status	Empirical Evidence & Theoretical Bounds
H1: Random Graphs HITS ~ In-Degree	**NOT SUPPORTED** (Attack Cost)	In minimal budget tests to penetrate Top-10 ($B^*(10)$ across n = 9 controlled trials on ER-500), HITS required a mean of **4.44× more budget** than simple in-degree (30–50 edges vs 5–10 edges). Clean rank correlation (τ = 0.746, ρ = 0.887 on ER-1000) is supplementary only.
H2: Spectral Phase Takeover at $p \approx \sigma_1^2$	**SUPPORTED** (Preliminary)	Empirical takeover occurs somewhat below σ_1^2 ($p^*(10) / \sigma_1^2$ = **0.817** [0.751, 0.879], n = 60). Bound $\sigma_1^2 \geq$ max in-degree holds in 60/60 instances (100.0%). Abrupt jump observed in 63.3% of curves (mean jump 69.7 ranks vs 16.9 for PageRank); 36.7% counterexamples (22/60).
H3: Dense Communities Cheaper Attachment	**INCONCLUSIVE / NOT SUPPORTED**	Evaluated A2 (isolated farm) vs proposal A5 (linking to target and c = 3 top real authorities, n = 18 trials). On ER control (n = 9), attaching saved **48.1%** [39.3%, 57.0%] budget. On Planted SBM (n = 9), target failed to reach Top-10 under A2 within $B \leq 200$, leaving cost saving undefined / NaN.
H4: Defense Pareto Trade-Off Frontier	**NOT SUPPORTED** (So Far)	Tested across 1,152 evaluations (n = 110 positive attack gain instances). **NO defense achieved ≥50% mean reduction at ≤5% clean NDCG loss across all attacks.** D4/D4b mitigate isolated farms but fail on camouflaged/gradient attacks; D2 destroys clean ranking (59.6% loss).

6. PATH TO FINAL SUBMISSION (20–25% REMAINING WORK)

Planned Technical Milestones:

- **WEBSPAM-UK2006 Host Graph Pipeline (Planned: 28 Oct):** Convert the compressed BVGraph bitstream to an ASCII edge list using Java WebGraph utilities, then evaluate baseline Spam@k, AUC, and precision/recall across rankers. *Note: Host graph size and format were not verified from the dataset page (the proposal assumed ~1.9 MB; per the proposal, to be confirmed).*
- **Experimental Scale-Up (Planned: 1 Nov):** Expand evaluations to 5 seeds (currently 3), budgets up to $B = 500$ (currently ≤ 100), additional target nodes, `web-NotreDame`, and higher budget cap ($B > 200$) for SBM community runs to resolve H3.
- **Defense Matrix on Larger Graphs (Planned: 5 Nov):** Run full defense matrix on large subgraphs with thresholds calibrated on a separate validation graph.
- **Experiments Freeze (Planned: 8 Nov):** Lock raw CSVs, regenerate 300 DPI figures, and conduct final hypothesis tests.
- **Final Submission (Planned: 15 Nov):** Final Report and presentation deck.

Status of Stretch Objectives:

- **Game-Theoretic Loop:** NOT STARTED. Iterative attacker-defender equilibrium model scheduled for final sprint.
- **Closed-Form Spectral Bound:** PARTIAL. Empirical takeover at $p^* \approx 0.82\ \sigma_1^2$ observed across 60 instances; formal closed-form mathematical proof and generalized $B^*(k)$ bound are in progress.
- **WEBSPAM-UK2007 Classifier:** NOT STARTED. Dependent on Milestone 1 host graph ingestion.

7. TEAM CONTRIBUTIONS

The project functional areas and contributions confirmed across all team members are detailed below:

Team Member	Contributions & Functional Areas
Shrisay Shandilya	Core sparse rankers (HITS, PageRank, SALSA), SNAP data loaders, Kleinberg base-set pipeline, clean computational efficiency baselines.
Vibhor Rathi	Adversarial link injection attacks (A1–A6, A4b, A5b), threat model validation suite, edge budget constraint verification.
Prachi Jain	Algorithmic defenses (D1–D5), evaluation metrics (NDCG, Jaccard, Tau, Rho), statistical testing suite (Bootstrap, Wilcoxon), publication figure generation.

8. OTHER INFORMATION & DISCLOSURES

Corrections vs Version 1 of Progress Report:

- **CPU Model:** Probed actual hardware as 12th Gen Intel Core i5-12500H (12 cores, 16 threads), correcting unverified v1 claim.
- **H1 Correlation:** Verified clean ER-1000 $\tau = 0.746$ ($p = 7.99 \times 10^{-244}$; $\rho = 0.887$), correcting unverified v1 claim of 0.912.
- **Defense Metric Bug:** Resolved metric defect that returned 1.0 on zero-gain cells, correctly assessing undefended reduction as 0.0%.
- **Defense D2 & D3:** Fixed D2 WCC no-op with true domain weighting; empirically verified D3 Hub Capping as an exact no-op on tested graphs.
- **Attack Definitions:** Restored proposal A4 (single fake page) and A5 (fake hubs with no cross-links), retaining multi-hub variants as A4b/A5b.

Verification, Reproduction & Ethics:

- **Automated Test Suite: 30 passed** unit tests in 11.4s under bare `pytest -v`, asserting ranker correctness vs NetworkX ($< 10^{-8}$).
- **Reproduction Commands:** Fully reproducible via relative scripts:
  - `python scripts/download_data.py`
  - `python scripts/run_clean_baselines.py`
  - `python scripts/run_tier_b_experiments.py`
  - `python scripts/compute_facts_v2.py`
  - `python scripts/generate_figures.py`
- **Research Ethics:** All experiments conducted offline on public, anonymized SNAP web datasets and synthetic benchmark topologies.
- **AI-Assistance Disclosure:** *Code and experiments were developed with an AI coding agent (Antigravity) under the team's direction; results were audited by the team.*